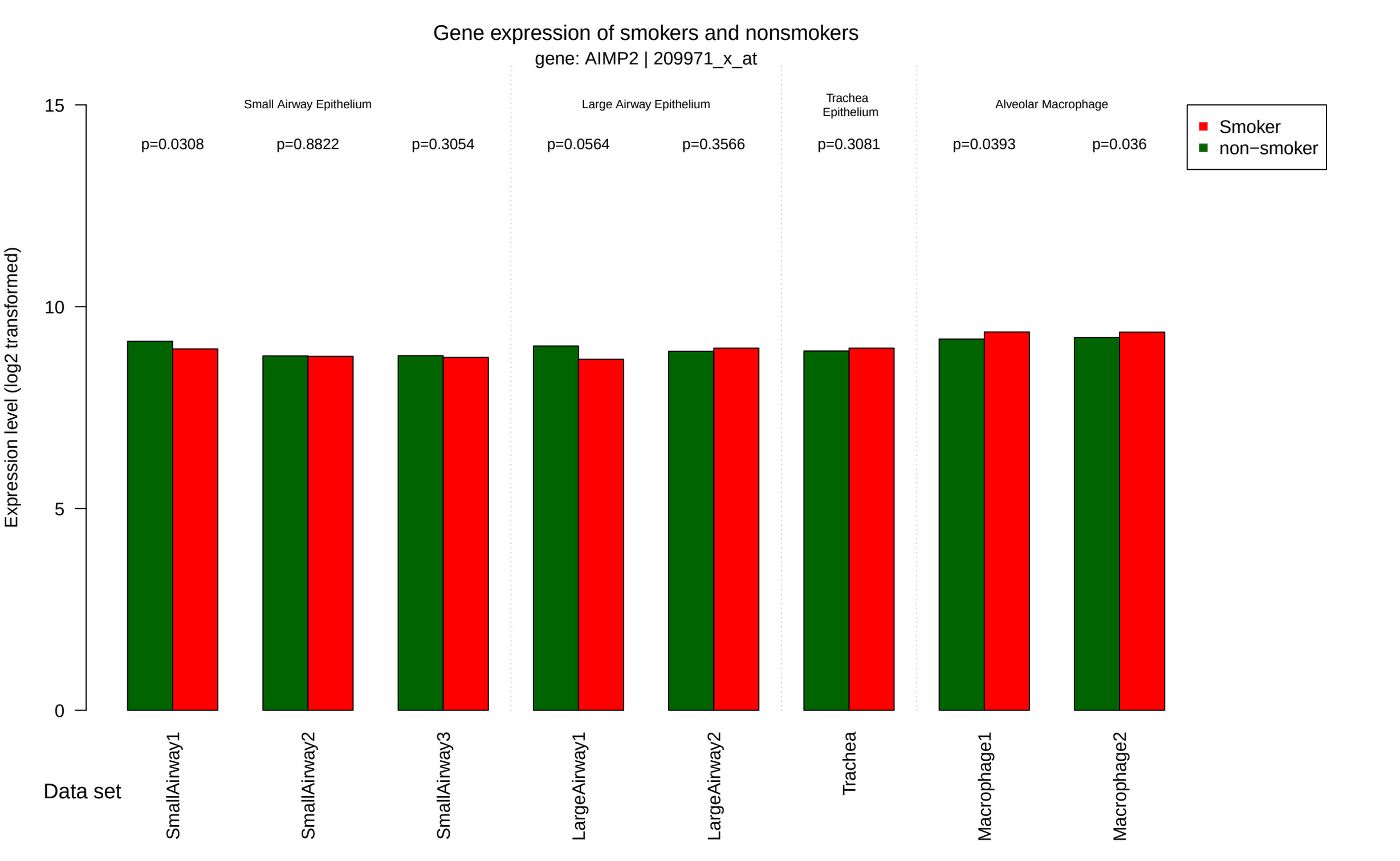
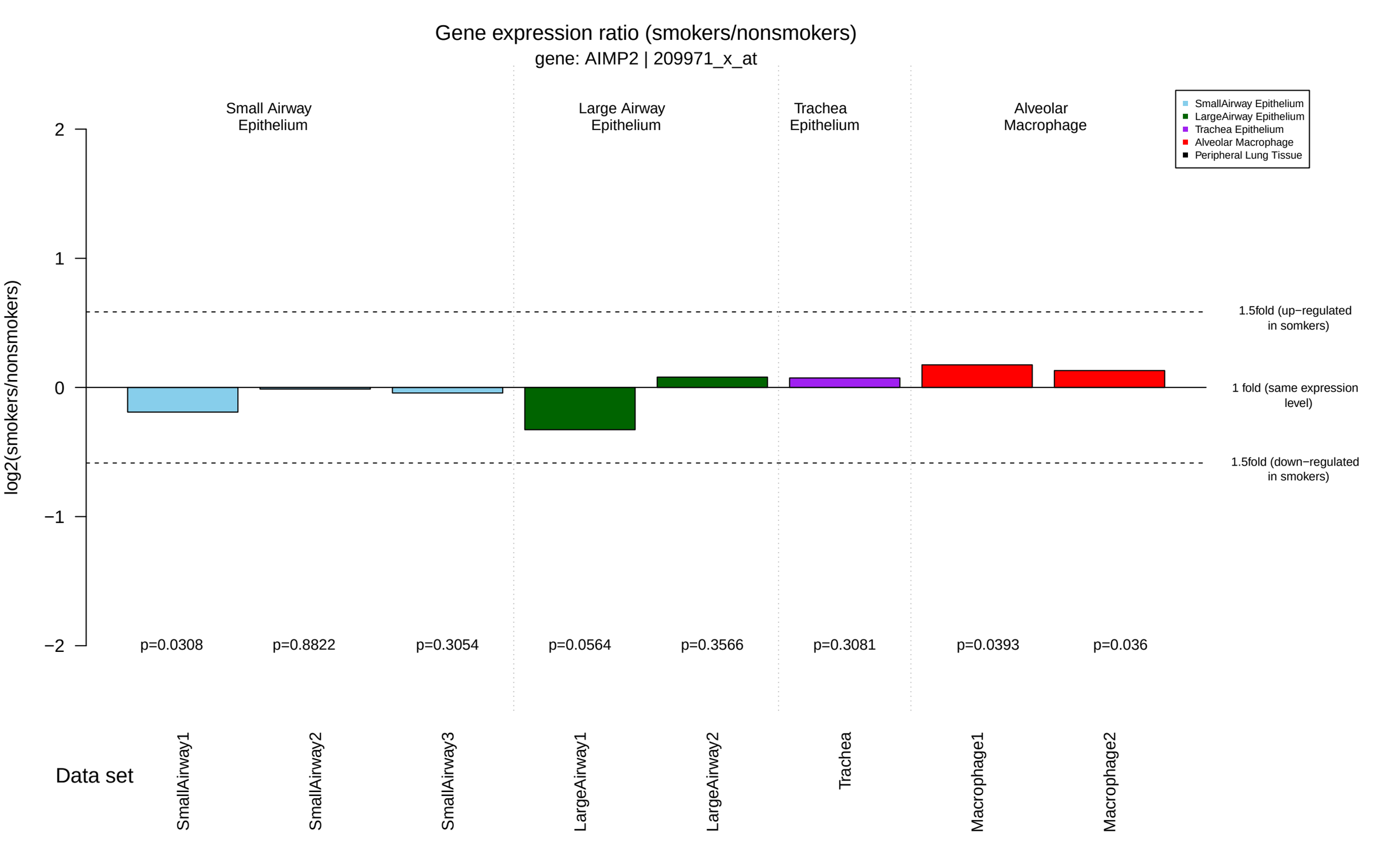
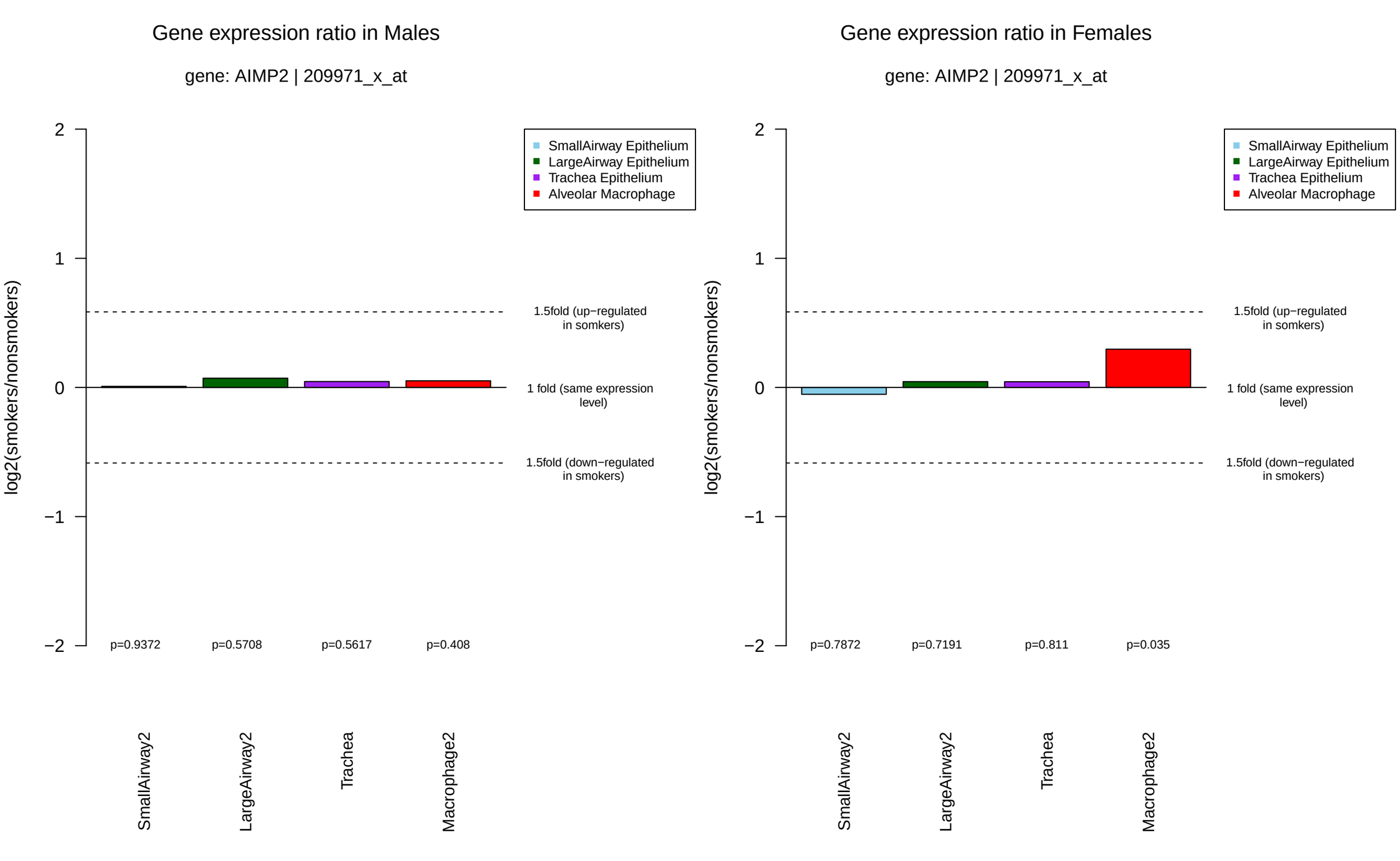
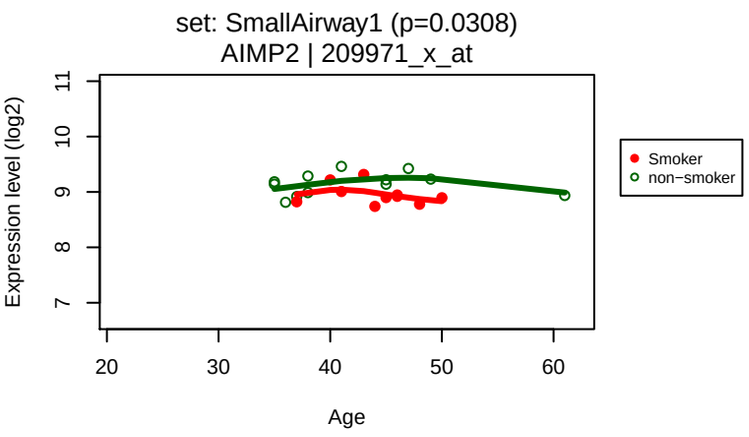
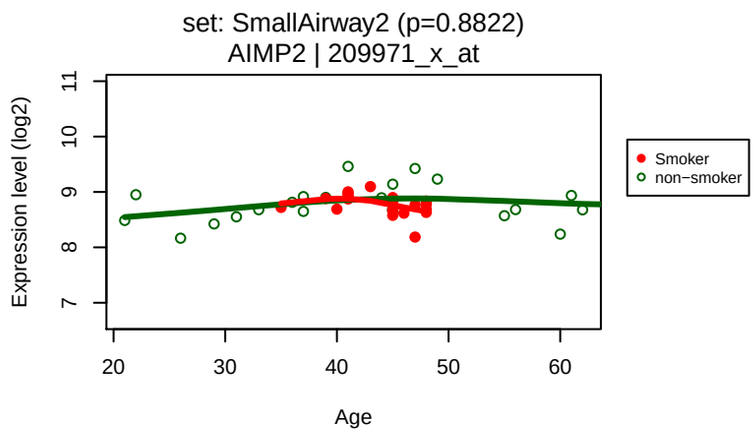
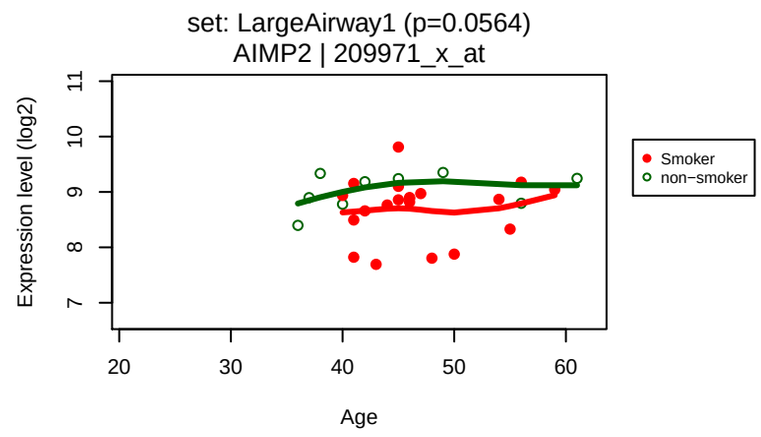

Gene expression of smokers and nonsmokers
gene: AIMP2 | 209971_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0308
p=0.8822
p=0.3054
p=0.0564
p=0.3566
p=0.3081
p=0.0393
p=0.036
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

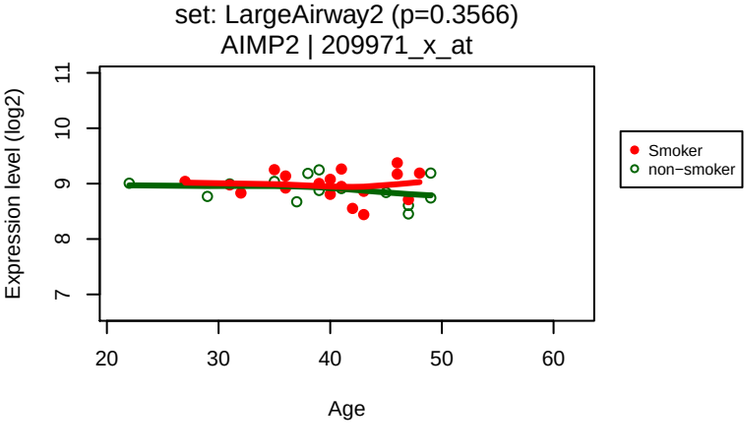

Gene expression ratio (smokers/nonsmokers)
gene: AIMP2 | 209971_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0308
p=0.8822
p=0.3054
p=0.0564
p=0.3566
p=0.3081
p=0.0393
p=0.036
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

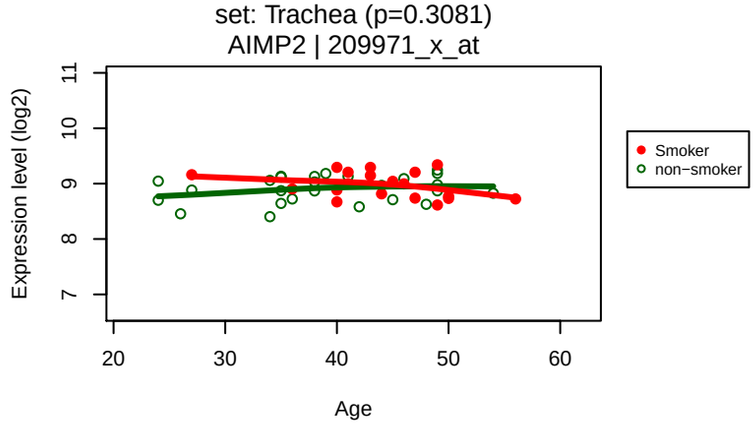

Gene expression ratio in Males
gene: AIMP2 | 209971_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9372
p=0.5708
p=0.5617
p=0.408
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AIMP2 | 209971_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7872
p=0.7191
p=0.811
p=0.035
SmallAirway2
LargeAirway2
Trachea
Macrophage2

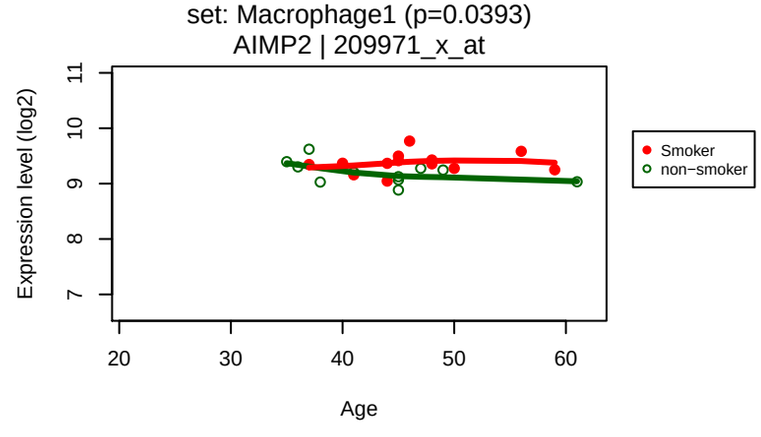

set: SmallAirway1 (p=0.0308)
AIMP2 | 209971_x_at
Expression level (log2)
Age
Smoker
non−smoker

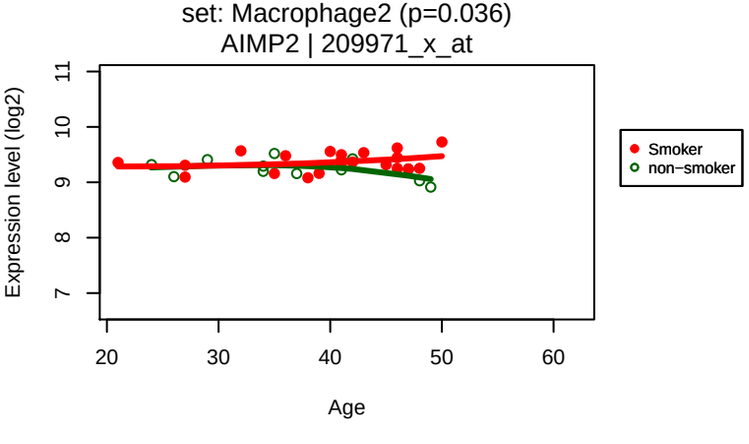

set: SmallAirway2 (p=0.8822)
AIMP2 | 209971_x_at
Expression level (log2)
Age
Smoker
non−smoker

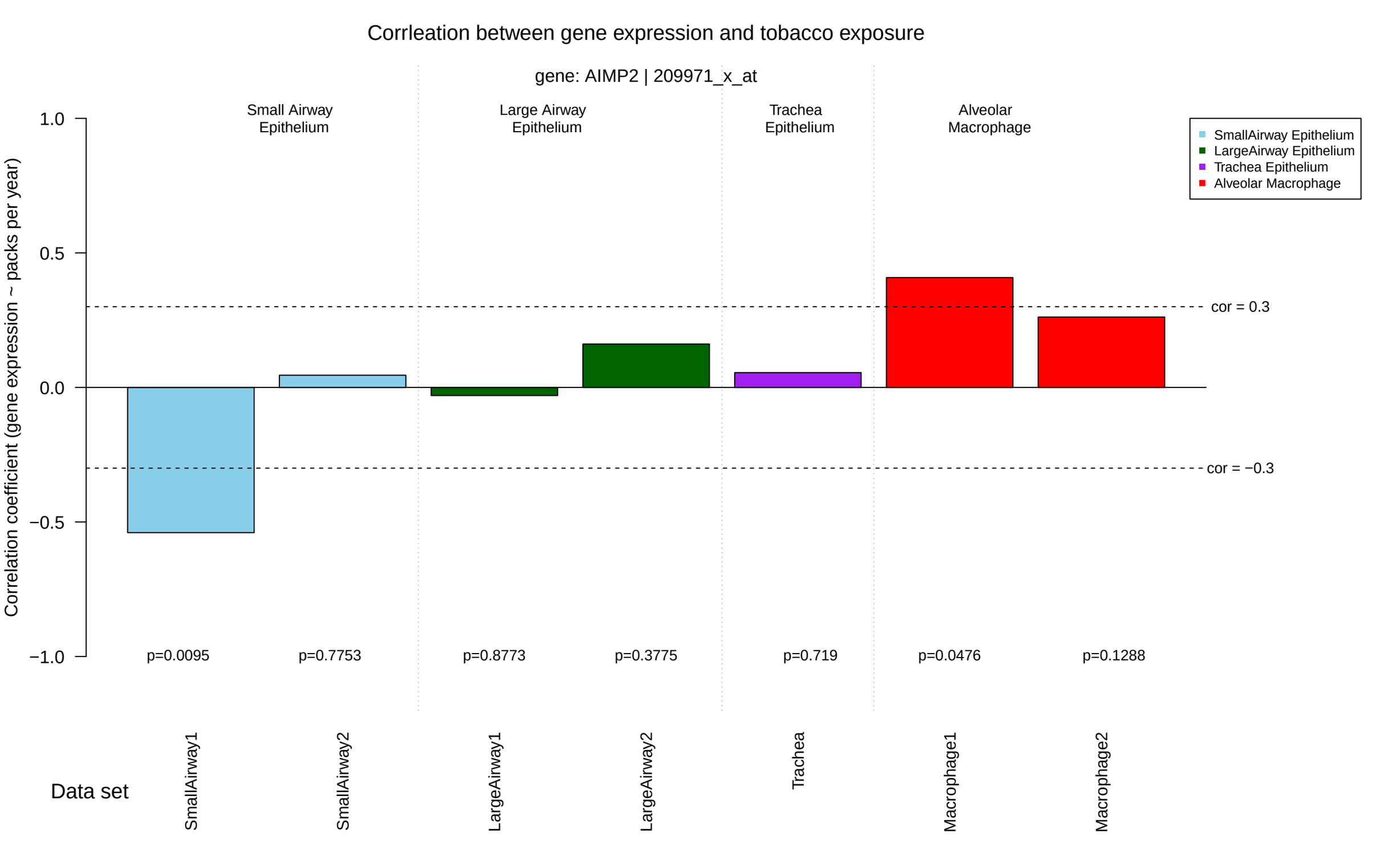

set: LargeAirway1 (p=0.0564)
AIMP2 | 209971_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3566)
AIMP2 | 209971_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3081)
AIMP2 | 209971_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0393)
AIMP2 | 209971_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.036)
AIMP2 | 209971_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AIMP2 | 209971_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0095
p=0.7753
p=0.8773
p=0.3775
p=0.719
p=0.0476
p=0.1288
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2